

UDN Group: Delivering engaging news and media services with smart analytics

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ABOUT UDN GROUP



By using Google Cloud, UDN Group can support smart analytics and host its news site. UDN Group is able to increase clickthrough rate on online ads and new content, enhance operational efficiency, and provide highly reliable digital media services.

About UDN Group

Established in 1951, United Daily News Group (UDN Group) is one of the largest newspaper media groups in Taiwan publishing United Daily News, Economic Daily News, and the World Journal. In 1999, UDN Group launched udn.com, its online news platform, and has since gradually expanded its business to digital advertising, publishing, ecommerce, and cultural events. Currently, udn.com has more than 16 million monthly unique visitors and attains 85 percent reach rate among people older than 15 years old in Taiwan.

Google Cloud results

- Enables seamless integration with the Google Marketing Platform for more efficient online advertising
- Boosts ML model development efficiency with Vertex AI
- Increases clickthrough rate by 2x for news content with analytics insights generated by BigQuery
- Zero downtime for news sites with highly reliable and

Enhanced
clickthrough
rate by 2x

Industries: Media & Entertainment

Location: Taiwan

Functions



About Merkle

Merkle is a global data-driven, tech-enabled performance marketing agency specializing in customer strategy, performance media, customer experience and personalization, customer relationship management, loyalty marketing and enterprise technology services. With more than 50 offices around the world, Merkle currently has 400+ world-class enterprise clients. It is a certified service partner for both Google Cloud and the Google Marketing Platform.

Google Cloud (<https://cloud.google.com/>)

enabled performance marketing agency

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

management, loyalty marketing and

Compute Engine

(<https://cloud.google.com/compute/>)

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Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

For more than 70 years, United Daily News Group (<https://www.udngroup.com/en>) (UDN Group) has been a leading watchdog in the Taiwanese media landscape that has received numerous journalism accolades. It publishes the United Daily News newspaper, which is among the largest traditional news dailies in the country. UDN Group also publishes Economic Daily News, which focuses on business news and the World Journal serving the global Chinese-speaking community.

As an industry leader, UDN Group started its digital transformation journey in 1999, when it launched udn.com, its online news platform. As more readers joined the digital world, the media group gradually expanded its services to online advertising, digital publishing, ecommerce, and digital news subscription. Currently, udn.com has more than 16 million monthly unique visitors, which represent two-thirds of Taiwan's population, and attains 85 percent reach rate among people above 15 years old in the country.

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"UDN Group has long been an influential news organization in Taiwan leading public discussions on wide-ranging topics," says Apollo Sun, COO at UDN Group. "As the media landscape shifts from print to digital, our services continue to expand and evolve, but providing engaging and impactful content remains our primary focus."

To enhance user engagement in a more structured way, UDN Group adopted a data-driven culture and created its data department in 2008. As traffic to its news websites grew, the group's on-prem infrastructure and data tools started to encounter scalability challenges. To obtain more cost-effective

scalability in the long run, UDN group in 2018 decided to migrate its online services and data analytics business to a cloud platform. It eventually landed on [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) because of its technology advancements and seamless integration with [Google Ad Manager](https://admanager.google.com/home/) (<https://admanager.google.com/home/>), which UDN Group had been using since 2013.

"Cloud migration is a must for us, because relying on on-prem resources to serve growing users and process ever-increasing data would have been too costly and time-consuming," explains Sun. "Besides providing data tools that can be seamlessly integrated with the online advertising platform, Google Cloud also met our needs when it came to cost, technology, and security compliance."

Enhancing ad clickthrough rate with ML models trained in Vertex AI

With technical support and training provided by the Google Cloud team, UDN Group completed the migration to Google Cloud seamlessly for its news sites and runs data analytics frequently. For online advertising, the group stores content data, audience tags, and user behavior data collected by [Google Analytics](https://marketingplatform.google.com/about/analytics/) (<https://marketingplatform.google.com/about/analytics/>) in [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>), which generates lists of target audiences that are then sent to Google Ad Manager for targeted advertising. Since BigQuery can be integrated with the other [Google Marketing Platform](#)

(<https://marketingplatform.google.com/about/>) tools, UDN Group is able to build a streamlined data pipeline to support efficient online advertising.

To achieve more accurate ad targeting, UDN Group kicked off a project in 2022 to build a user preference prediction machine learning (ML) model together with Merkle (<https://www.merkle.com/>), a Google Cloud and the Google Marketing Platform partner. With its marketing expertise, Merkle helped UDN Group complete the development of the model within eight months.

"Although we have our own data specialists, we believe that partnering with external teams can help us identify our blind spots and further improve our services," notes Anson Mok, Data General Manager at UDN Group. "With its rich experience in online marketing, Merkle not only provides technical know-how, but also fully understands our needs as an ad publisher."

For this project, UDN Group and Merkle used Vertex AI (<https://cloud.google.com/vertex-ai>) to run algorithms that extract keywords in UDN Group's online content for tagging, match content tags and ad categories, and generate target audiences. During the development process, the joint team conducted a series of A/B testing to select the best algorithms. Mok shares that the highly convenient development environment that Vertex AI provides helped the team explore data and conduct model building more efficiently. Leveraging the new model, UDN Group has enhanced the unique clickthrough rate for certain ad categories by up to four times. With the

outstanding results, the project won the Digital Singularity Award for Best Precise Marketing Application (<https://udn.com/news/story/7238/7594225>) presented by the Digital Marketing Association in Taiwan in November 2023.

"Our online marketing business has greatly benefited from the adoption of Google Cloud, because it enables smooth workflow from data collection, storage, analytics to ad display, and provides tools like Vertex AI that can effectively help us improve our ad performance," adds Mok. "Currently, 40 percent of our ad orders are supported by our data-driven advertising services."

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—Anson Mok, Data
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Providing more engaging news content with BigQuery

Besides online advertising, UDN Group also uses data insights generated by BigQuery to produce more engaging news content, empower its online recommendation engines, and improve its operational efficiency. Now, 10 to 15 percent of the news content published on udn.com is created and designed based on user preference data. On average, this is double the conversion rate compared to other content.

"We rely on data insights in almost all aspects of our operations. BigQuery is cost-effective and helpful for us to increase reader engagement and optimize our online services to attract more users," says Mok.

Enabling zero-downtime news services with diverse infrastructure tools

UDN Group has been leveraging the cloud infrastructure of Google Cloud to deploy its news sites, apps, and other online services for more than five years. Depending on the nature of its products, it

employs virtual machines (VMs) in Compute Engine (<https://cloud.google.com/products/compute?hl=en>), Kubernetes containers in Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) or serverless architectures in Cloud Functions (<https://cloud.google.com/functions>) to support computing. On top of the resource diversity, these computing tools also enable high scalability that accommodates fluctuating traffic.

"Using Google Cloud to run our digital services means that we have diverse computing options to run different products in the most suitable and cost-effective way," notes Mok. "At the same time, we're able to maintain our role as a reliable news source by ensuring high service stability. We've never experienced any downtime after migrating to Google Cloud."

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Increasing media services' width and quality with the power of data

With its well structured and functioning data pipelines hosted in Google Cloud, UDN Group is ready to achieve more by leveraging the power of data. It plans to launch a digital reading platform in 2024, providing personalized services through data analytics and exploring more ML model applications with Merkle. The group will also be tapping on the latest generative artificial intelligence (AI) technology of Google Cloud to enhance its operational efficiency.

"After seeing the benefits brought by data analytics, we're convinced that all our digital services in the future will be deeply linked with data," says Sun.

"With its powerful data and infrastructure tools, Google Cloud has helped us effectively improve our services and has become an integral part of our operations. We'll continue working with Google Cloud to deliver products and services that meet our customers' expectations."